

Feature Engineering And Exploratory Data Analysis (EDA)

Data Science Life Cycle (ML Life Cycle)

1. Collecting of Data
2. EDA (Analysis)
3. Processing (Pre – Processing)
4. Model
5. Evaluate and Validate Model

Statistics:- Statistics is the science of collecting, organising and analyzing the data.

Collecting, Organise, Interpretation, Analysis -----> Insight

Examples:- Scientific Problem, Healthcare, Social Problem.

Problems Statements:-

Sales of Product -----> Sales is going down

Analysis -----> Product, Paying to Customer, Leadership, Marketing, Competitor.

Data Set -----> Analysis -----> Conclusion

- Project Manager
- Business Analyst (Domain Expert)
- Data Scientists

Only Domain require :- **EDA & FE** -----> Used to get conclusion from the dataset.

Collecting of Data:-

Examples: Big Data Tools, SQL, NO-SQL, Website, Webscraping, Some files format: (CSV, XML, JSON)

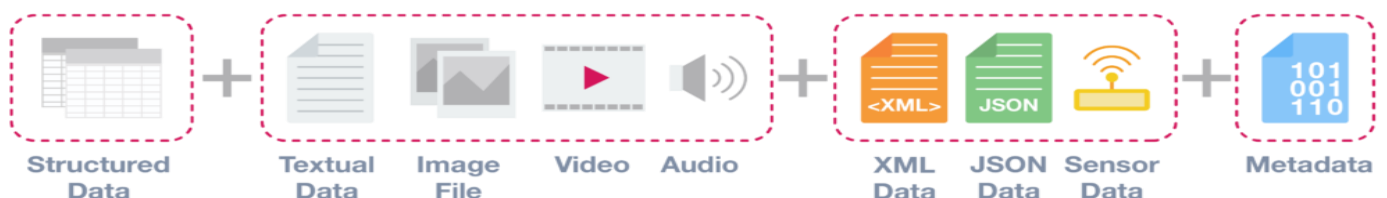
Types Of Data:-

Batch Data: Historical Data, Mini batch Data (Periodic)

Streaming Data: Continuous Data (Live Data)

Data

- 1) **Structure Data :** Table (Row Column) -----> ML
- 2) **Unstructure Data:** Video, Images, Voice, Sound, Text -----> DL
- 3) **Semi Structure Data:** JSON, XML.



Structure Data			
Quantitative (Numerical)		Qualitative (Categorical)	
Continuous	Discrete	Ordinal	Nominal
Consists of numerical values that can be measured but not counted.	Consists of numerical values that can be counted.	Consists of text or labels that have a logical order.	Consists of text or labels that have no logical order.
e.g. Weight, Height, Age, Speed {56.06 Kg, 87 Kg}	e.g. whole no. pin code, No. of children {0, 1, 2, 3}	e.g. Beverage size {small, medium, large}, 10 th , 12 th , Grad, UG, PG, Phd	e.g. Profession {chemist, carpenter}, Male, Female

Students Performance (Dataset)

F --> Feature

Name (F1)	Age (F2)	Height (F3)	Sex (F4)	Weight (F5)	Education (F6)
Nikhil	20	183	Male	65	PG
Prakriti	19	158	Female	-	UG
Categorical	Numeric	Numeric	Categorical	Numeric	Categorical
Nominal	Continue	Continue	Nominal	Continue	Ordinal

Used to find types of data:

Univariate: Single Column

Bivariate: Two Column

Multivariate: More than two columns

Independent And Dependent Variable:

Age, (Height, Sex) ----> Independent

Weight ----> Dependent

Q. First EDA is required or FE or Pre-Processing.

Ans. EDA is Required (Analysis of the data based on the given data).

EDA ----> Pre-processing ----> Model

EDA (Analysis):

1. Profile of the Data
2. Statistic Analysis
3. Graph Based Analysis

Profile of the Data:-

Analysis

- 1) Row
- 2) Column
- 3) Missing
- 4) Category
- 5) Numeric
- 6) Duplicate
- 7) DType
- 8) RAM (D Size)

Statistics Analysis (Interpretation):-

Univariate, Bivariate, Multivariate

- 1) Variance
- 2) Covariance
- 3) Standard deviation
- 4) Correlation
- 5) Chi Square Test
- 6) T-test
- 7) Z-test
- 8) Anova test
- 9) Mean, Median, Mode

Graph Based Analysis (Plotting):-

- 1) Box Plot ---> Outlier, Distribution, Statistical Profile.
- 2) Scatter Plot ---> Outlier, Linear or not
- 3) Pie Chart
- 4) Histogram ---> Distribution
- 5) KDE Plot
- 6) Count Bar ---> Used for counting (Bar Chart)
- 7) Heat Map ---> Correlation

Q. Based on EDA, can we do a pre-processing of data? ----> Ans. Yes

Pre-Processing of data:-

- 1) Missing Value Handel
- 2) Outlier Handel
- 3) Scaling of Data
- 4) Transformation (Log, Box-cox, Square, Cube)
- 5) Encoding
- 6) Imbalance Data
- 7) Feature Selection
- 8) Dimension Reduction (PCA, LDA, TSNE)
- 9) Duplicate Value / Duplicate Column
- 10) Split / Merge / Drop / Add (column)

These are steps of Feature Engineering:

1. Missing Null Values ----> Missing value handel
2. Outlier ----> Handel
3. Categorical ----> Encoding (to change categorical data into numerical data)
4. Skewed Range ----> Scale (Within a certain range)
5. Count of Feature ----> Handel Imbalanced data, Feature Selection, Dimension Reduction.

Some Automated Tools In Python For EDA:-

1. Pandas Profiling
2. Sweet Viz
3. AutoViz
4. D-Tale
5. Mito
6. Knime
7. Dataprep

Ways of Performing Feature Engineering:

1. Missing Value Handel

- I. Fill with Random number.
- II. Forward/Backward Filling.
- III. Statistical Approach (Mean, Median, Mode).
- IV. With the help of end of distribution, fill the Missing Values.
- V. Drop the row.
- VI. Impute with KNN (KNN-Imputer)
- VII. ML Algorithm for Missing Values.
- VIII. Build own ML Model to predict Missing Values.

2. Outlier Handling

Detect the Outlier

- I. Z-Score
- II. IQR
- III. Box-Plot
- IV. Scatter Plot
- V. Violin Plot

Handel Outlier

- I. Drop
- II. Fill with Median/Mean/Mode
- III. Replace / Trimming

3. Transformation

- I. Box Cox Transformation
- II. Power Transformation
- III. Log
- IV. Square
- V. Cube
- VI. Yeo Johnson

4. Scaling of Data

- I. Standardization
- II. Min Max Scalar
- III. Unit Scaling

5. Encoding Various Methods

- I. One Hot Encoding
- II. Label Encoding
- III. Binary Encoding
- IV. Target Guided Encoding
- V. Hash Encoding

6. Imbalanced Dataset Treatment Various Methods

- I. Collect More Data
- II. Under Sampling
- III. Over Sampling
- IV. Cluster Based Over Sampling

How to find the best Model Accuracy Various Methods:

- a. To increase the accuracy, We need to change the pre-processing technique and use different method or steps from the above.
- b. We needs to use each and every pre-processing steps and find the best accuracy.

Q. How do we transform the data?

Import numpy as np

np.log(df)

sns.displot(df)